Stage 0 on receipts: the quotient probe law and the probe distribution (2026-10-15)

Status: measured on archived receipts (zero compute, no timed run). This is stage 0 of two uncommitted crypto-autoresearcher proposals:

- `IDEA-20261015-7a3c9e`: folded Koblitz index calculus is baby-step giant-step on the quotient. It predicts probes per target `r/(A·#states)` and `IC_online/BL_online = c_probe·c_state/(2.1·c_step²)`, independent of `r`.
- `IDEA-20261015-b41f68` (stages 0–1 here, B1–B2): the per-target probe count is geometric, and IC ns/probe is the host's random-access latency over the table's bytes.

Each verdict applies the proposal's own falsification conditions as written. Every row uses one experiment's receipts. Prices from different experiments are never combined into one row. `analysis.json` holds every figure; `stage0.py` recomputes it.

Verdicts

idea	criterion (as written in the proposal)	result	verdict
7a3c9e	prediction 1 outside [0.7, 1.3] at any size → falsified	0.93–1.13 at the five m=2 §23 sizes; 1.02 at n=61 K500; 0.88–1.09 at the four n=83 dense-root bases; **2.30** at n=83 K600 (16 points, 95% [0.66, 4.27]), and **1.40** on the K600 rank stage (mean of 600 relations, run 38523707244)	**falsified at n=83 K600**; holds at the other 10 rows
7a3c9e	slope of ln(IC/BL) on ln r > 0.25 → falsified	six §23 sizes: 0.086 [−0.11, 0.28] (§23 model column), 0.139 [−0.03, 0.31] (c_BL=2.1, tuned step).	not falsified; this tests only a model, because no BL ran at §23
7a3c9e	formula off by more than 10× → falsified (prediction: within 3×)	Neither BL value was measured the one measured IC/BL ratio, n=83 K600 run 38689817618: formula 94.6 against 235 [61, 645] measured, a ratio of 0.40	inside 3×
7a3c9e	Koblitz vs ordinary curve	no ordinary-curve receipt exists	not tested
b41f68	probe CV < 0.6 or > 1.5 → falsified	all 64-point panels: 0.80–1.01; n=61 W=256: 0.95; **n=83 K600 16-point panel: 1.57** [0.95, 2.22]; KS against exponential p = 0.001	**falsified on the n=83 K600 panel**; holds on the 64-point panels and at n=61

Overall:

- **7a3c9e is falsified by its own rule 1.** The quotient probe law `r/(A·#states)` is accurate to 15% on 10 of 11 receipt rows. On B's n=83 K600 configuration, which carries the 235× BL measurement, the mean is 2.3× the law. The rank stage's 600 relations put it at 1.40×.
- **The price product lands inside its factor-3 band at the only size with a measured BL.** The proposal's own back-of-envelope (231 against 235) agreed only by accident. Its two inferred prices are wrong:
  - c_state is 18.4 μs, not 2.1 μs: P_IC is 549.8 s over 2.99e7 states, not 2^28;
  - c_step is 180 ns, not 40 ns.

  The two errors cancel (Figure 2).
- **b41f68 fails its CV rule on the one panel the claim is about,** n=83 K600. It passes everywhere else.

Table A: probe law and price product, one experiment per row

Definitions:

- r/(A·#states): `A = 2n` signed-Frobenius classes; #states = stored folded pair states (§23 `stored_pairs`) or root-table entries (four-summand root index).
- **c_probe** (measured): mean IC online ns over mean probes, on the same rows.
- **c_state** (measured, by definition): P / #states. Each row's P is stated in `analysis.json`.
- **c_step** (measured): mean walk ns over mean steps for that experiment's rho.
- **formula**: c_probe·c_state/(2.1·c_step²).
- **IC/BL_model**: measured IC online over 2.1·(r/A)·c_step²/P. This is a model, because no BL ran in those experiments.

row (source)	design	$\log_2$ r	N	probes measured / predicted [95%]	c_probe ns	c_state ns	c_step ns	formula	IC / BL
§23 k0n41	m=2 pair table	39.0	64	28,412 / 25,233 = **1.13** [0.89, 1.39]	43.3	375	112	0.61	model 0.69 (§23 col 2.5)
§23 k0n57	m=2	38.0	64	11,062 / 10,183 = **1.09** [0.85, 1.34]	59.4	544	155	0.64	model 0.70 (3.3)
§23 k0n53	m=2	44.3	64	56,384 / 53,541 = **1.05** [0.83, 1.31]	69.7	279	124	0.60	model 0.63 (2.0)
§23 k1n59	m=2	44.5	64	51,156 / 54,968 = **0.93** [0.74, 1.13]	93.7	352	130	0.93	model 0.86 (2.8)
§23 k0n61	m=2	47.2	64	218,824 / 213,078 = **1.03** [0.79, 1.29]	90.8	921	129	2.38	model 2.45 (8.0)
§23 k1n47	**m=3** (outside the claim)	36.6	64	4.3 / 7,572 (not the m=2 law)	—	—	—	—	—
n=61 a=0 K500, GHA 37156392134	four-summand root index	47.2	256	89,110 / 87,563 = **1.02** [0.90, 1.14]	298	1,444	227	3.96	model 4.03
n=83 K1271 regular, dense-root panel	root index, 8 GiB	52.9	64	371,728 / 385,040 = **0.97** [0.79, 1.16]	462	1,242	175[1]	8.9	model 8.6
n=83 K1271 dense	root index, 4 GiB	52.9	64	same probes = **0.97**	408	1,294	175[1]	8.2	model 8.0
n=83 K1500 dense	8 GiB	52.9	64	300,022 / 276,452 = **1.09** [0.85, 1.34]	431	1,256	175[1]	8.4	model 9.2
n=83 K1798 dense	8 GiB	52.9	64	169,070 / 192,403 = **0.88** [0.69, 1.08]	454	1,442	175[1]	10.2	model 9.0
n=83 K600, BL arm 38689817618	root index, 4 GiB[2]	52.9	16	3,979,401 / 1,727,838 = **2.30** [0.66, 4.27]	349	18,403	180 (rT770)	**94.6**	**measured 235 [59, 649]**[3]; model 218
n=83 K600, Tier 1 38523707244 (B)	same	52.9	16	same probes (work-identical)	575	29,526	707 (rung 3, not tuned)	16.2	model 37.3
n=83 K600, reconcile 38617911080	same	52.9	16	same probes	511	25,543	280 (rT770)	79.0	model 182

Notes:

1. The dense-root panel's c_step comes from that study's single primary-point rho run (`runs/rho_primary`), on the same shared M4 Pro host. One walk, unisolated: this price is weak.
2. 4 GiB is derived from the source, not recorded: `RootTable128` has 2^27 slots of 32 bytes (`protocol.json` of `../ic-probe-latency-20261015`).
3. The bootstrap interval here (paired over 16 targets, this script's own seed) is [58.9, 649]. The run's own frozen figure is [61, 645].

Readings, labelled:

- **Measured.** At the §23 sizes, IC online over a BL model priced at the tuned step and `c_BL = 2.1` is 0.63–2.45. The §23 column's 2.0–8.0 priced the walk at the cheaper canonical step with 1.93·1.21. Neither is a measurement of BL.
- **Derived.** At those rows, formula ÷ (IC/BL model) equals predicted probes ÷ measured probes by algebra. So prediction 2 adds nothing to prediction 1 there. The proposal's claim (4), that the formula "reproduces the 2.0–8.0×", cannot be tested without a BL arm at those sizes.
- **Measured.** At n=83 K600 (BL arm), formula ÷ measured = 0.40. It splits into two factors:
  - predicted ÷ measured probes, 1/2.30 = 0.43;
  - measured ÷ modelled BL online, 0.93. BL online steps averaged 31,506 against 35,473 from `2.1·(r/A)/P_steps`.
- **Measured.** Across the three K600 runs (same work, three EPYC runners), c_probe runs 349–575 ns. Against the tuned walk, c_probe ÷ c_step is 1.94 (run 38689817618) and 1.82 (run 38617911080).

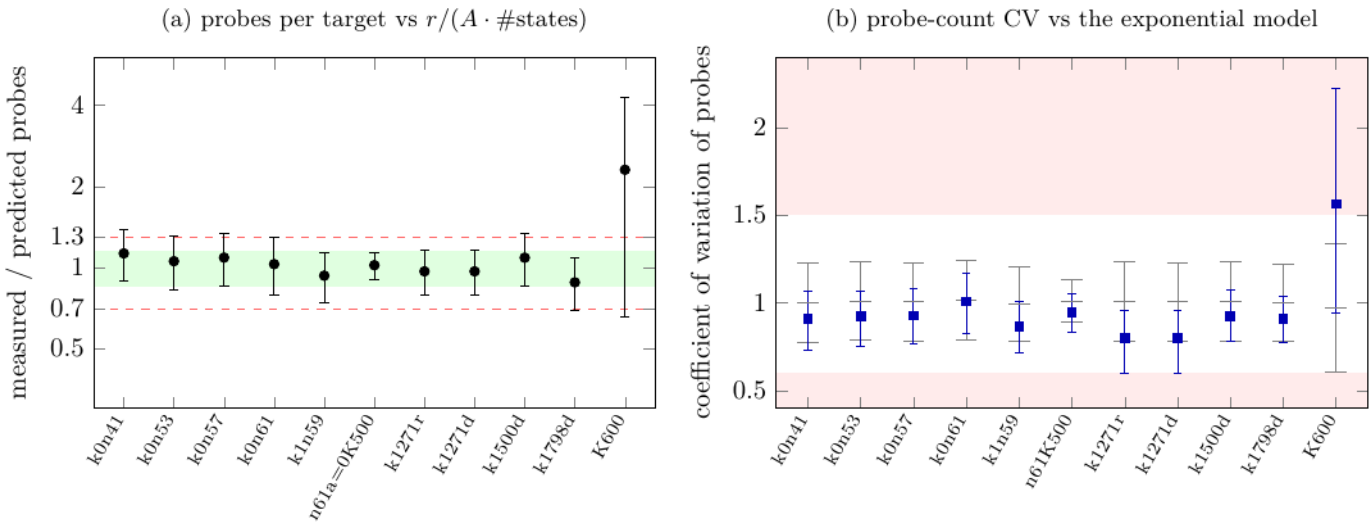


Figure 1: Figure 1

Figure 1. (a) Mean probes per target over $r/(A \cdot \#\text{states})$, one row per experiment, 95% bootstrap over targets. Green: the claimed ±15% band. Dashed: the [0.7, 1.3] falsification band. The m=3 §23 size is omitted, and K600 is plotted once because its counts are identical in its three runs. (b) The sample CV of probes with a 95% bootstrap interval (blue). Grey: the 95% range of the CV of N exponential draws. Red: the b41f68 falsification region. Source `figures/probe_law.tex`*, data* `probe_ratio.dat` *and* `probe_cv.dat`*, vector* `probe_law.pdf`*.*

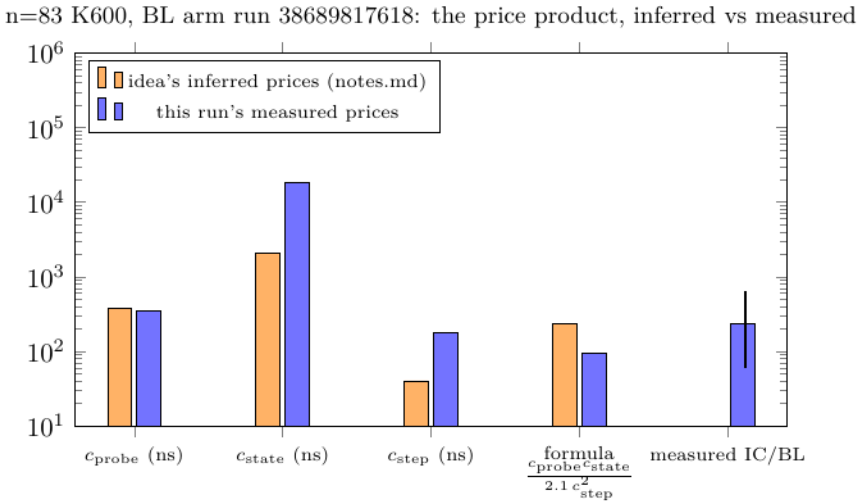


Figure 2: Figure 2

Figure 2. n=83 K600, run 38689817618: the three prices and the formula as `IDEA-20261015-7a3c9e.notes.md` *inferred them (orange) and as this run measured them (blue), with the measured IC/BL ratio and its interval. Source* `figures/price_product.tex`*.*

Table B1: probe-count distribution against the exponential (geometric) model

The KS test is Lilliefors-type: it uses the exponential with the sample mean, and its p-value comes from 2,000 parametric bootstrap draws (seeded). The family is 12 panels, so the Bonferroni level is 0.05/12 = 0.0042. The smallest attainable p is 1/2001, below that level.

panel	N	CV [95% boot]	exp-null CV 95%	median/mean	max/mean (≤ ln N + 1?)	KS D	p	CV in [0.6, 1.5]
§23 k0n41	64	0.909 [0.73, 1.07]	[0.77, 1.23]	0.705	4.34 (≤ 5.16)	0.097	0.35	yes
§23 k0n53	64	0.925 [0.75, 1.07]	[0.79, 1.23]	0.689	4.32	0.101	0.29	yes
§23 k0n57	64	0.929 [0.77, 1.08]	[0.78, 1.23]	0.714	3.71	0.072	0.75	yes
§23 k0n61	64	1.008 [0.82, 1.17]	[0.79, 1.24]	0.591	4.57	0.076	0.69	yes
§23 k1n59	64	0.867 [0.71, 1.01]	[0.78, 1.21]	0.799	3.86	0.101	0.26	yes
§23 k1n47 (m=3)	64	0.807	[0.78, 1.23]	0.928	4.41	0.207	0.0005	yes (exponential rejected)
n=61 K500, W=256 (isolated hosted run)	256	0.946 [0.83, 1.05]	[0.89, 1.13]	0.756	6.31 (≤ 6.55)	0.055	0.20	yes
n=83 K1271 regular / dense	64	0.801 [0.60, 0.96]	[0.78, 1.24]	0.905	4.08	0.154	0.011 / 0.018	yes
n=83 K1500 dense	64	0.925 [0.78, 1.08]	[0.78, 1.24]	0.627	3.61	0.066	0.83	yes
n=83 K1798 dense	64	0.911 [0.78, 1.04]	[0.78, 1.22]	0.587	3.70	0.090	0.45	yes
n=83 K600, 16 fresh points	16	**1.567** [0.95, 2.22]	[0.60, 1.34]	0.283	4.55 (> 3.77)	0.385	**0.001**	**no**
(context) n=83 BL online steps, 16 points	16	1.209	[0.60, 1.40]	0.725	4.60	0.268	0.034	—

- **The m=2 §23 panels, n=61 and the dense-root bases.** None is rejected at the Bonferroni level. The CV of 7 of 10 in-scope panels lies inside the proposal's tighter [0.9, 1.1]. k1n59 (0.87) and K1271 (0.80) fall below it, so they are lighter-tailed than exponential.
- **n=83 K600 is heavier-tailed than any exponential of its size.**
  - Three of 16 fresh points took 14.5M–18.1M probes, against a law mean of 1.73M, and Q0 took 8.85M.
  - For an Exp(1.73M), P(X > 14.4M) = 2.4e−4.
- **Hypothesis, not tested.** The cause may be the wide producer's scan order. It is source-read from crypto `770f6a69` `examples/koblitz_orbit_dlp_fast_online.rs`, wide `extract`:
  - The scan is left-orbit-major. One left orbit is 600·83·332 ≈ 16.5M probes.
  - With only 600 columns, about 1.6 three-sum representations are expected per (target, left orbit). So about 20% of targets would find no hit in their first block and need another full block.
  - That predicts the tail at about one block length, and the excess mean.
  - At K1798 the same estimate gives about 40 per block, which is why the dense bases look geometric.
  - Testing it needs the per-block hit counts or a re-run with a shuffled scan order; neither is done here.

B2: IC ns per probe against table bytes

row	host	table bytes	c_probe ns (all online)	lookup-only ns
§23, five m=2 sizes	x86_64, crypto §23 host (`runs/host.json`); 2.4e5–6.3e6 stored pairs	not recorded	43–94	22–50
n=61 K500, GHA 37156392134	EPYC 7763	not recorded (peak RSS 1.45 GB, 1.5e7 entries)	298	—
n=83 K1271 regular / dense, same host	Apple M4 Pro, shared	8 GiB / 4 GiB (allocated)	462 / 408	—
n=83 K1500 / K1798 dense	same	8 GiB / 8 GiB	431 / 454	—
n=83 K600 runs 38689817618 / 38617911080 / 38523707244	EPYC 9V45 / 7763 / 9V74	4 GiB (derived)	349 / 511 / 575	≈ online (PDP is 99.99%)

- **Same host, same probes, half the bytes.** K1271 at 4 GiB (dense) against 8 GiB (regular) is 12% cheaper per probe. This is unisolated, one study.
- **Across hosts at fixed work.** K600 spans 1.65× (349–575 ns) over three EPYC runners.
- **The pointer-chase null object does not exist yet in any receipt.** No run has measured it, so b41f68's predictions 2 and 3 remain untested here. Stage 1, the pointer chase, and stage 2, the huge-page pair, are prepared as a Tier 1 workflow in `../ic-probe-latency-20261015/`.

Sources (all read-only; one experiment per row)

The crypto commits are crypto `ecc27ef8` (local `main` checkout); the files read were last changed at `30745078` and `0f0db951`:

- `research/ic_single_target_20260930/runs/k*n*/T??-R1.price.json` and `analysis.json`;
- `research/n83_dense_root_20261013/analysis_panel.json`, `runs/panel/*/targets_result.jsonl` and `runs/rho_primary/receipt.json`;
- `research/sat_factor_base_review_20260908/autolab/evidence/20261003-koblitz-n61-hosted-isolated/autolab_ru`

cryptanalysis sources:

- `main` `0382c341`:
  - `experiments/koblitz-single-target-n83-tier1-20261014/tier1_github_38523707244/`;
  - `experiments/koblitz-n83-reconcile-20261014/run_github_38617911080/`.
- branch `claude/n83-bl-arm` `4b8ff92a` (PR #918):
  - `experiments/koblitz-n83-bl-arm-20261014/run_github_38689817618/`.

Not used. The 2026-10-12 isolated n=61 K500 64-point panel (autolab run `20261012T173628Z-3fd988ea23`) is not committed: its receipts are on a Modal volume, and a GitHub code search found nothing. The n=61 hosted isolated run `gha-37156392134-1` (W=256, EPYC 7763, `isolated_bench reserve`) stands in for it.

```
python3 -I stage0.py --crypto /path/crypto \
  --n83-tier1 ../koblitz-single-target-n83-tier1-20261014/tier1_github_38523707244 \
  --reconcile ../koblitz-n83-reconcile-20261014/run_github_38617911080 \
  --bl-arm <checkout of
  ↪  claude/n83-bl-arm>/experiments/koblitz-n83-bl-arm-20261014/run_github_38689817618 \
  --out analysis.json
python3 -I make_figures.py analysis.json figures
(cd figures && pdflatex probe_law.tex && pdflatex price_product.tex)
```

Claim classes and limits (controlled-experiment skill)

- **Exact computation.** Probe and step counts, r, #states, and table geometry read from source.
- **Comparative / statistical.**
  - Every ratio carries a bootstrap or t interval.
  - The KS tests are Bonferroni-corrected over 12 panels.
  - The detector's power at N=16: an exponential sample of 16 has a CV above 1.34 with probability 2.5%.
  - Receipts already exist, so this is a retrospective analysis; nothing was preregistered beyond the proposals' own thresholds.
- **Heuristic.** BL online $2.1 \cdot (r/A) \cdot c_step^2/P$ is a model wherever no BL arm ran. The scan-order explanation for K600 is a hypothesis.
- **Out of scope.**
  - No speed claim for any method.
  - No transfer to ECC2K-130.
  - The four-summand root index and the m=2 pair table are different designs. The proposal's claim names the "folded pair-state" index calculus, and its own n=83 example is the four-summand root index, so both were tested.

Success criterion (AGENTS.md). This is a negative/attribution result, not an IC baseline improvement. The next improvement it names is the stage 2 huge-page pair on B's K600 producer. If that pair lowers ns per probe on the same instance with every answer verified, it is an online-stage engineering improvement.